

BLSN Report BLSN Synthetic Validation Report Theme: light Vehicle P3 Orion
Cooling Mode helium_closed_loop Nominal Mass Flow 110 g/s Limits 80 - 140 g/s
Sanity Check Telemetry Status Message PASS [PASS] PASS | Nominal mass flow is
bounded by configured limits Tuple Execution Telemetry Targets (seconds) Metric
Target Warn Threshold Fail Threshold ci_total_runtime_s 120.0 180.0 240.0
pytest_runtime_s 15.0 25.0 35.0 pipeline_runtime_s 5.0 10.0 15.0
yaml_json_lint_runtime_s 3.0 6.0 9.0 Metric Markdown Slugs Ci Total Runtime S
Pytest Runtime S Pipeline Runtime S Yaml Json Lint Runtime S Ci Total Runtime S
Metric ID: `ci\_total\_runtime\_s` Target Threshold: ****120.0s**** Warn Threshold:
****180.0s**** Fail Threshold: ****240.0s**** Pytest Runtime S Metric ID:
`pytest\_runtime\_s` Target Threshold: ****15.0s**** Warn Threshold: ****25.0s**** Fail
Threshold: ****35.0s**** Pipeline Runtime S Metric ID: `pipeline\_runtime\_s` Target
Threshold: ****5.0s**** Warn Threshold: ****10.0s**** Fail Threshold: ****15.0s**** Yaml
Json Lint Runtime S Metric ID: `yaml\_json\_lint\_runtime\_s` Target Threshold:
****3.0s**** Warn Threshold: ****6.0s**** Fail Threshold: ****9.0s**** System Verification:
Claimed vs. Actual Claimed Limits: 0 | Actually Tested Limits: 0 -> STATUS:
UNKNOWN Execution Velocity Total Runtime (engine + tests): 0.0000s Validation
Strictness Pass: 0.00% | Fail: 0.00% | Skip: 0.00% Wave 12: SPC & Telemetry
Stats Runtime Drift Current Runtime: 0.0000s Previous Moving Average: 0.0000s
Historical Control Total Accumulated Runs: 1 Success Rate: 100.00% DoE Factors -
A: python_version , B: os_type , C: load_parameter Principal Component
Eigenvalue Explained Variance Highest Weight Variable PC1 0.000000 0.00%
total_tests PC2 0.000000 0.00% skip_rate Interactive Plotly Visualizations
System Priority & Bottleneck Ranking (Bradley-Terry) Pairwise probability
model: $P(i \text{ \> } j) = p_i / (p_i + p_j)$ Global Repo Ranking Rank Artifact BT
Strength Win Prob vs Top No global BT ranking data. Inter-Category Ranking Rank
Category W
in Prob vs Top No inter-category BT ranking data. QPLANT Requirements Ranking
Bridge Rank Requirement Win Prob vs Top No QPLANT requirement ranking data.
Mermaid System Flow Diagram graph LR; SSOT[SSOT Config] --> PIPELINE[Pipeline
Execution]; PIPELINE --> MARKDOWN[Markdown Slug Pages]; PIPELINE --> INDEX[Index
HTML]; PIPELINE --> SLIDES[Slides HTML]; INDEX --> PDF[PDF Export]; (function ()
{ const emptyLayout = { xaxis: { visible: false }, yaxis: { visible: false },
annotations: [{ text: 'No Plotly payload available', showarrow: false, xref:

```
'paper', yref: 'paper', x: 0.5, y: 0.5 }}]; const renderFallback = () => {  
  Plotly.newPlot('plotly-telemetry-chart', [], emptyLayout);  
  Plotly.newPlot('plotly-pca-chart', [], emptyLayout); };  
  fetch('visuals/plotly_payloads.json') .then((response) => response.ok ?  
  response.json() : Promise.reject(new Error('Missing Plotly payload')))  
  .then((payload) => { const charts = payload && payload.charts ? payload.charts :  
  {} ; const telemetry = charts.telemetry_combo || {} ; const pca =  
  charts.pca_bottlenecks || {} ; Plotly.newPlot('plotly-telemetry-chart',  
  telemetry.data || [], telemetry.layout || emptyLayout); Plotly.newPlot('plotly-  
  pca-chart', pca.data || [], pca.layout || emptyLayout); }) .catch(() =>  
  renderFallback()); })();
```